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Section 4.3
Some Rules For Probability

Addition Law:-

Is used to Find Prob that atleast one of two events occurs. In simple, it means Event A happens or Event B happens OR both happen.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- $P(A)$ = Prob of A
- $P(B)$ = Prob of B
- $P(A \cap B)$ = Prob that both A and B happen
- $P(A \cup B)$ = Prob that A or B or both happen.

Why we do substract $P(A \cap B)$?
Because when we add $P(A) + P(B)$ the common part (where both events happen) is counted twice, so we substract it once.

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Special Addition Law:

If two events cannot happen together they are mutually exclusive events.

$$P(A \cap B) = 0$$

So Law Becomes.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B) - 0$$

$$P(A \cup B) = P(A) + P(B)$$

Addition Law For Three Events:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

If mutually Exclusive.

$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

Compliment Rule:

States that prob of an Event occurs equals to 1 minus prob that it does Not occur.

$$P(A) = 1 - P(A^c)$$

↓ prob that the event occurs.

↓ prob that the event doesn't occur.

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Example: 4.10

QS pg 162.

$P(DUEUF) = P(D) + P(E) + P(F)$

$P(DUEUF) = 0.149 + 0.077 + 0.106$

$P(DUEUF) = 0.332$

$P(DUEUF) = 33.2\%$

Example: 4.11

QS pg 163 Data 162.

a-

Let, event L = Less that 200 acres

$P(L) = 1 - P(I)$

$P(L) = 1 - 0.035$

$P(L) = 0.965$

$P(L) = 96.5\%$

b-

~~56~~ 8

let event M: 50 acres or more

First Find under 50 acres or more

$P(U) = 0.084 + 0.265$

$P(U) = 0.349$

$\therefore P(M) = 1 - P(U)$

$P(M) = 1 - 0.349 = 0.651$

Date: Day:

Example : 4.12.

Given Data:

Total Card = 52

Spade card = 13

Face card = 12. Common Card = 3

a:

Let event

S : selecting Spade

F : selecting Face.

Solution:-

$$P(S \cup F) = \frac{f}{N} = \frac{22}{52} = 0.4231$$

$$P(S \cup F) = 42.31\%$$

b-

$$\therefore P(S \cup F) = P(S) + P(F) - P(S \cap F)$$

$$P(S \cup F) = \frac{13}{52} + \frac{12}{52} - \frac{3}{52}$$

$$P(S \cup F) = \frac{22}{52} = 0.4231$$

$$P(S \cup F) = 42.31\%$$

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Example: 4.13

Given Data:-

Male : 76.2% = 0.762

Under 18 = 15.3% = 0.153

Male and Under 18 = 10.8% = 0.108

Let events:

M : Person is male

E : Person is Under 18

$P(M \cup E) = P(M) + P(E) - P(M \cap E)$

$P(M \cup E) = 0.762 + 0.153 - 0.108$

$P(M \cup E) = 0.806$

$P(M \cup E) = 80.6\%$.

Exercise: 4.3.

Question: 4.67.

Qs on Pg 166.

Given Data:

Total Poker = 12

Red = 3 White = 4

Blue = 5

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Find:

Prob that blue is selected.

Let event:

B: Poker chip selected is Blue.

$$P(B) = \frac{f}{N} = \frac{5}{12} = 0.4167$$

$$P(B) = 41.67\%$$

2nd Method

$$P(B) = \frac{\binom{5}{1}\binom{3}{0}\binom{4}{0}}{\binom{12}{1}} = \frac{(5)(1)(1)}{12} = \frac{5}{12}$$

$$P(B) = 0.4167$$

$$P(B) = 41.67\%.$$

Question: 4.69

Q:s Pg 166

Let events:

A: senator is under 50

B: senator in his or her 50

C: Senator is in his or her 60

D: Senator is under 70.

Date:_____ Day:_____

a:

$$P(S) = \frac{f}{N} = \frac{12+33+32}{100} = \frac{77}{100}$$

$P(S) = 0.77$

$P(S) = 77\%$

b-

$S = A \cup B \cup C$

c-

P(A):-

$$P(A) = \frac{12}{100} = 0.12 = 12\%$$

P(B):-

$$P(B) = \frac{33}{100} = 0.33 = 33\%$$

P(C):-

$$P(C) = \frac{32}{100} = 0.32 = 32\%$$

d-

$P(S) = P(A) + P(B) + P(C)$

$P(S) = 0.12 + 0.33 + 0.32$

$P(S) = 0.77$

$P(S) = 77\%$

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Comparison: Answer calculated using special addition rule (0.77) is identical to answer obtain using f/N rule in part (a) (0.77)

Question: 4.61

Qs Pg 166-167

a-

Smoked

Let event S: Trader who smoked

$P(S) = 1 - P(A)$

$P(S) = 1 - 0.733$

$P(S) = 0.267$

$P(S) = 26.7\ \%$

b-

Let event H: smoked atleast 1 per day

$P(H) = P(C) + P(D) + P(E) + P(F) + P(G_1)$

$P(H) = 0.078 + 0.053 + 0.028 + 0.007 + 0.003$

$P(H) = 0.169$

$P(H) = 16.9\ \%$

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Question: 4.77 QS pg.

a-

Let event A: laborers Lived in US in b/w 1 and 20 yrs.

Let event B: Laborer Lived in US less than 11 years

$$P(A \cup B) = 0.17 + 0.30 + 0.21 + 0.12 + 0.13$$

$$P(A \cup B) = 0.93$$

$$P(A \cup B) = 93\%$$

b

Using General Addition Rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = (0.30 + 0.21 + 0.12 + 0.13) + (0.17 + 0.30 + 0.21 + 0.12) - (0.30 + 0.21 + 0.12)$$

$$P(A \cup B) = 0.76 + 0.8 - 0.63$$

$$P(A \cup B) = 0.93$$

$$P(A \cup B) = 93\%$$

c-

Part (a) is easier coz we can Directly add percentages of all categories included in either event

Date: _____ Day: _____

Question: 4.79

events:-

A: sum of dice is 7

B: Sum of Dice is 11

C: sum of Dice is 2

D: Sum of Dice is 3

E: Sum of Dice is 12

F: Sum of Dice is 8

G = event doubles or Rolled.

a:

Solution:

$P(A) = 6/36$ $P(B) = 2/36$

$P(C) = 1/36$ $P(D) = 2/36$

$P(E) = 1/36$ $P(F) = 5/36$

$P(G) = 6/36$ $\{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$.

b.

$P(A \cup B) = P(A) + P(B)$

$P(A \cup B) = 6/36 + 2/36$

$P(A \cup B) = 8/36$

$P(A \cup B) = 0.2223$

$P(A \cup B) = 22.23\%$.

$A \cap B = 0$ coz sum of 7 and 11 can't occur at same roll

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c-

$P(C \cup D \cup E) = P(C) + P(D) + P(E)$

$P(C \cup D \cup E) = \frac{1}{36} + \frac{2}{36} + \frac{1}{36}$

$P(C \cup D \cup E) = \frac{4}{36}$

$P(C \cup D \cup E) = 0.1112$

$P(C \cup D \cup E) = 11.12\%$

d-

$P(F \cup G_1) = \frac{10}{36} = 0.2778$

$P(F \cup G_1) = 27.78\%$.

e-

$P(F \cup G_1) = P(F) + P(G_1) - P(F \cap G_1)$

$P(F \cup G_1) = \frac{5}{36} + \frac{6}{36} - \frac{1}{36}$

$P(F \cup G_1) = \frac{10}{36} = 0.2778$

$P(F \cup G_1) = 27.78\%$.

Date: _____ Day: _____

Question : 4.81

Given:

Std who attend public school = 84.8%

std who attend college = 23%

Std who attend both = 17.7%

Event:

C: attend college

S: attend Public

$P(S \cup C) = P(S) + P(C) - P(S \cap C)$

$P(S \cup C) = 0.848 + 0.23 - 0.177$

$P(S \cup C) = 0.901$

$P(S \cup C) = 90.1\%$.

Question : 4.83

Given:

$P(A) = 1/4$

$P(B) = 1/3$

$P(A \cup B) = 1/2$

If Two event are mutually exclusive then

$P(A \cup B) = P(A) + P(B)$

$1/2 = 1/4 + 1/3$

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$\frac{1}{2} = \frac{7}{12}$

$\frac{6}{12} \neq \frac{7}{12}$

So not mutually exclusive.

b-

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$P(A \cap B) = \frac{1}{4} + \frac{1}{3} - \frac{1}{2}$

$P(A \cap B) = \frac{7}{12} - \frac{1}{2} = 0.0833$

$P(A \cap B) = 8.33\%$

Question : 4.85

Let events:

T: Getting Times

H: Getting Herad

E: Getting Examiner

Given

$P(T) = 47\% = 0.470$

$P(H) = 33.4\% = 0.334$

$P(E) = 34.6\% = 0.346$

$P(T \cap H) = 11.9\% = 0.119$

$P(T \cap E) = 15.1\% = 0.151$

$P(H \cap E) = 10.4\% = 0.104$

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$P(T \cap H \cap E) = 4.8\% = 0.048$

$P(T \cup H \cup E) = P(T) + P(H) + P(E) - P(T \cap H)$
$- P(T \cap E) - P(H \cap E)$
$+ P(T \cap H \cap E)$

$P(T \cup H \cup E) = 0.470 + 0.334 + 0.346$
$- 0.119 - 0.151 - 0.104 + 0.048$

$P(T \cup H \cup E) = 0.824$

$P(T \cup H \cup E) = 82.4\%$